

7.9 FLIGHT CONTROL SYSTEM (see Fig. 7-13)

The flight control system comprises a cyclic stick, a collective lever, and directional pedals to produce control inputs to the main rotor hydraulic servo actuators and tail rotor pitch change links.

Cyclic and collective servo actuator control inputs are combined by mixing levers and further transmitted to a swashplate. The swashplate changes the linear motion from the mixing levers to rotating motion and conveys pitch changes to the four main rotor blades.

Yaw control inputs are transmitted to the tail rotor sliding sleeve. The pitch of each blade changes when directional control inputs cause the nonrotating portion of the sliding sleeve to act upon the rotating portion. Pitch links between the sliding sleeve rotating portion and the pitch horn of each blade cause collective pitch angle changes of the tail rotor blades.

7.9.1 Cyclic Stick (see Fig. 7-13 and 7-14)

The cyclic stick is used to induce helicopter movements about the pitch and roll axes. The basic cyclic stick grip has switches for RPM-warning audio cutoff, cyclic control force trimming, and radio and internal communications microphone keying.

NOTE Instead of the new cyclic stick centering device (without locking possibility) in some H/C's a cyclic stick locking device is installed (observe ASB MBB-BO105-40-106). This locking device has a preset breaking point. It provides the possibility to override the locking device in case of emergency by a strong jerky force-aft movement of the cyclic stick (e.g. unintended locked cyclic stick in takeoff situation).

7.9.2 Collective Lever (see Fig. 7-13 and 7-14)

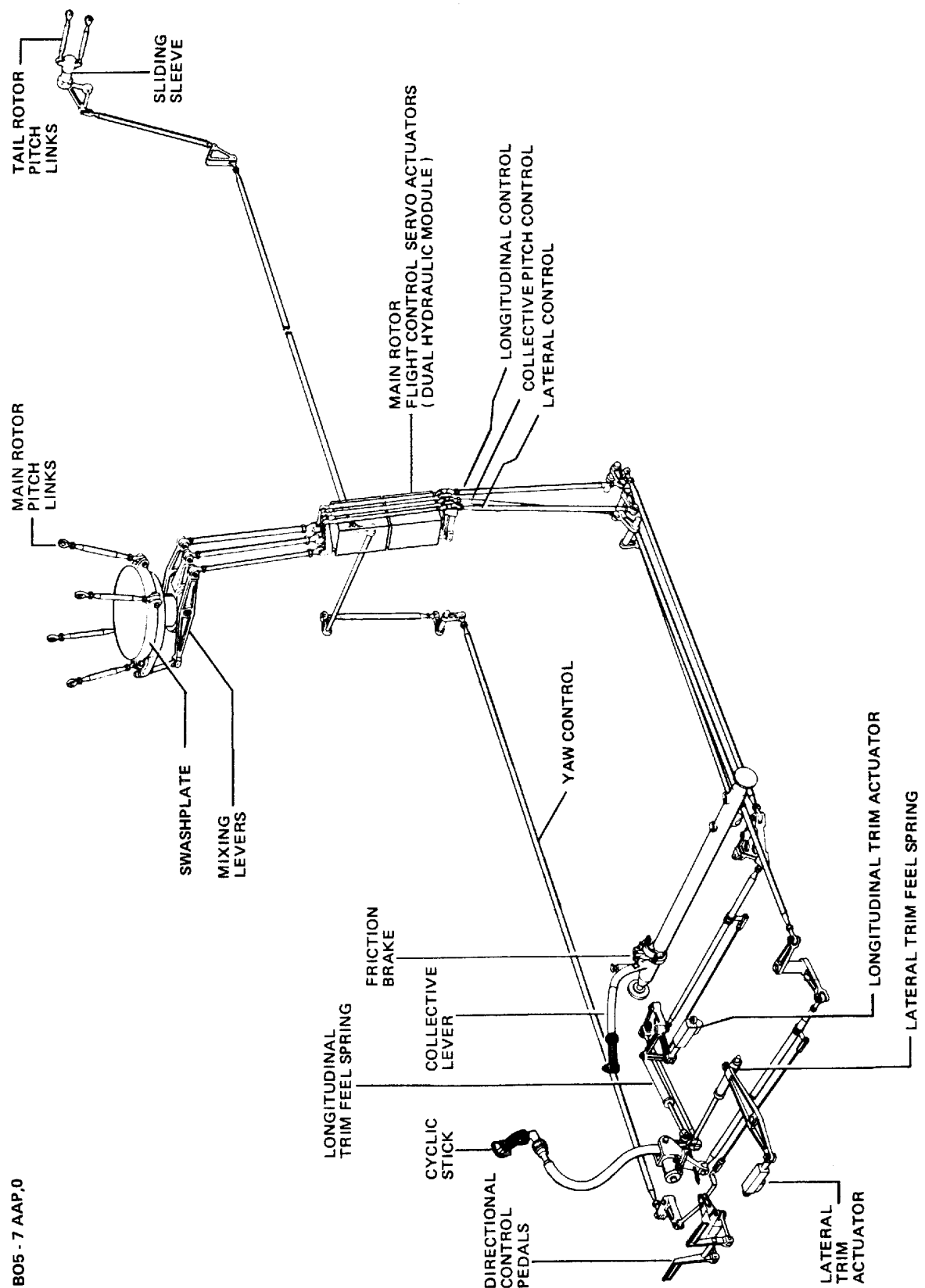
The collective lever provides a means of adjusting pitch angle of the main rotor blades and fuel flow metering requirements of the power turbine. A switch panel at the end of the lever has several switches installed, however, only the 4-way ENGINE TRIM switch and, on S-751 and subsequent, the LANDING LIGHT switches are active on the standard model. A friction adjustment at the base of the lever is used to prevent the collective pitch setting from moving when the lever is released.

7.9.3 Directional Control Pedals (see Fig. 7-13)

The directional control pedals are used to induce helicopter movement about the yaw axis. The pedals are adjustable fore-aft by use of a knob located aft of and between the pedals.

7.9.4 Cyclic Trim System (see Fig. 7-13 and 7-14)

The position of the cyclic stick may be electrically trimmed using the cyclic trim system. A lateral and a longitudinal trim feel spring assembly, each with an electrical actuator, are incorporated to produce artificial feel and cyclic control system trim. The spring assemblies will hold the cyclic stick in a neutral-feel trimmed position. Movement of the cyclic stick will cause the spring assemblies to compress and provide feel to the controls. When control pressure is released the cyclic will return to the neutral-feel position.



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Fig. 7-13 Flight Control System

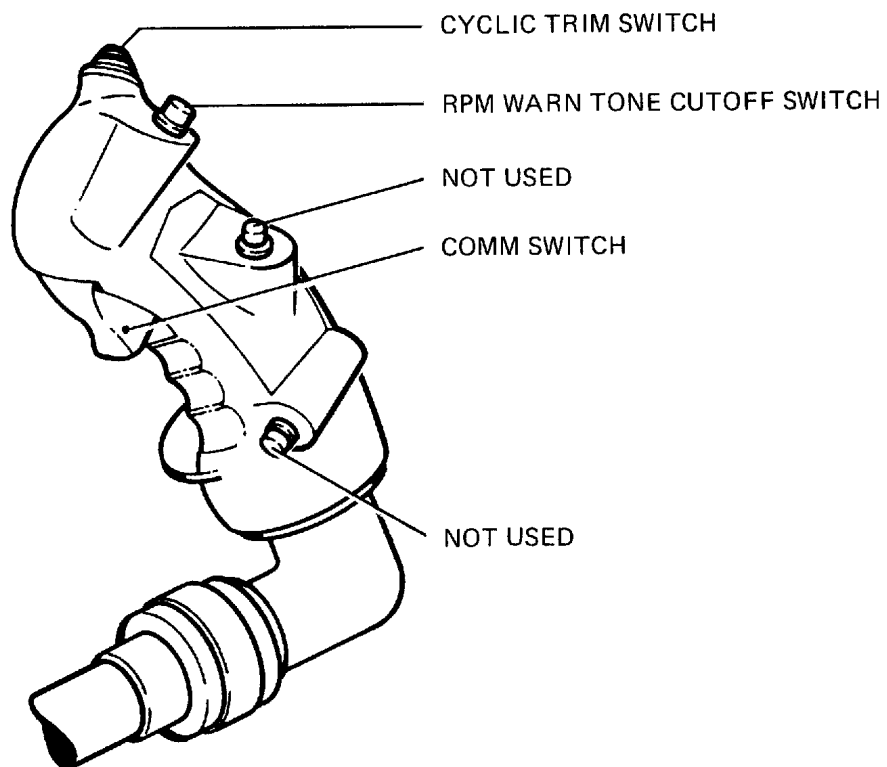
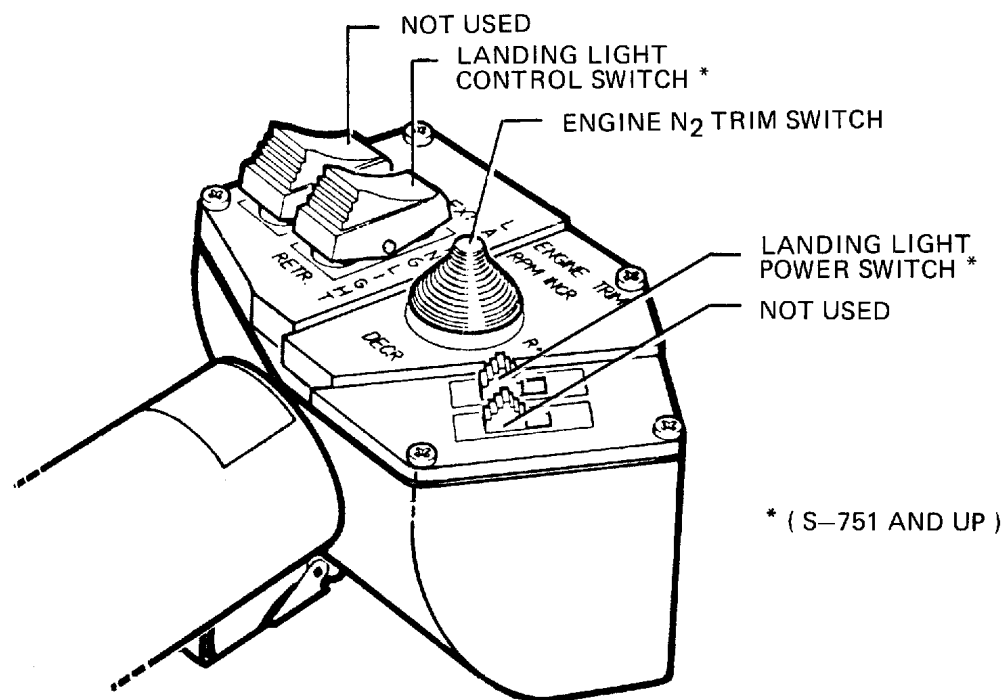
Manual retrimming of the neutral-feel position is accomplished by use of the 4-way TRIM switch on top of the cyclic stick grip. Pressing the switch in the desired direction will activate the appropriate trim actuator motor to reposition its respective spring assembly to establish a new neutral-feel position.

The system is powered direct from the Main Bus via the TRIM ACTUATOR circuit breaker on the center console switch panel.

7.9.5 **Flight Control Servos** (see Fig. 7-13 and 7-16)

The flight control servos are utilized to amplify the flight control forces applied to the main rotor system and thereby ease pilot workload.

Each of the three main rotor control axes has a tandem hydraulic servo actuator comprised of a primary servo and a standby servo. These servos are independent of each other to provide redundancy in the flight control system. Should one hydraulic system fail the remaining system can drive the main rotor flight control servos.



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Fig. 7-14 Cyclic Stick Grip and Collective Lever Controls